

In-Context Learning: Chain-of-Thought vs. Tree-of-Thoughts Prompting on Multi-Step Reasoning

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Abstract. We present an empirical comparison of four In-Context Learning (ICL) prompting strategies on the GSM8K grade-school mathematics benchmark: Zero-Shot Chain-of-Thought (CoT), Few-Shot CoT (8-shot), Tree-of-Thoughts (ToT) with BFS exploration at two search depths, and Self-Consistency voting. All experiments use Llama 3.1 8B via the Groq API without fine-tuning. Zero-Shot CoT achieves **95.0%** accuracy, Self-Consistency **92.0%**, and ToT (BFS-2, depth=6) together with Few-Shot CoT each reach **85.0%**—all on the same 100-problem seed-42 sample. After an answer-extraction format fix, Zero-Shot CoT alone sits on the Pareto frontier of accuracy and inference cost at the 8B scale. A depth ablation (depth=3 vs. depth=6) yields only +4.0 pp at $2.1\times$ inference cost, indicating diminishing returns when the evaluator is the same 8B backbone. A cross-model replication on Claude Haiku 4.5 with full evaluation identity preserved collapses the inter-strategy gap from 14 pp to 2 pp, revealing a *ceiling effect*: deliberation- and ensemble-style strategies pay off only when the base model has headroom to lose and identifiable wrong branches to prune. A pairwise Cohen’s κ analysis confirms this picture independently—ToT depths on Llama reach near-zero agreement ($\kappa=0.011$) versus perfect agreement on Haiku ($\kappa=1.000$).

Keywords: In-Context Learning · Chain-of-Thought · Tree-of-Thoughts · GSM8K · Large Language Models

1 Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities in natural language tasks through In-Context Learning (ICL), where the model adapts to new tasks from demonstrations provided in the prompt without updating model weights [1]. This ability to generalise from a handful of examples has made ICL the dominant paradigm for practical LLM deployment, motivating active research into how the structure and content of prompts influence multi-step reasoning quality. Among ICL techniques, Chain-of-Thought (CoT) prompting has emerged as a particularly effective approach, guiding models to produce

intermediate reasoning steps before arriving at a final answer [5]. The zero-shot variant [3] further simplified this by showing that a single trigger phrase suffices for instruction-tuned models to produce step-by-step reasoning, suggesting that structured reasoning has been internalised during pre-training.

Tree-of-Thoughts (ToT) extends this paradigm by exploring multiple reasoning branches simultaneously and pruning low-quality paths through a separate evaluation step [6]. While ToT has shown impressive gains over CoT on GPT-4, its effectiveness on smaller, open-weight models remains unexplored. A central open question is whether the multi-branch search ToT provides translates to accuracy gains when the evaluator model is the same small backbone—or whether evaluator weakness undermines the pruning mechanism entirely.

This paper presents a controlled empirical comparison of four prompting strategies (Zero-Shot CoT, Few-Shot CoT (8-shot), ToT with BFS, and Self-Consistency) on the GSM8K benchmark [2], using Llama 3.1 8B as a common backbone. Zero-Shot CoT reaches 95.0% on 100 samples (after an answer-extraction format fix, Section 4), our ToT implementation (depth=6, BFS with branching factor 2) reaches 85.0%, and Self-Consistency voting reaches 92.0%. A depth ablation (depth=3 vs. depth=6) further shows that doubling the search depth yields only +4.0 pp accuracy at $2.1\times$ the inference cost, suggesting that the bottleneck for ToT at the 8B scale is evaluator discrimination rather than search depth. To test whether these findings are model-specific, we replicate all five configurations on Claude Haiku 4.5, a frontier-tier model, with exact sample, prompt, and evaluation identity preserved—a controlled comparison that reveals a scale-dependent ceiling effect.

2 Related Work

The foundational prompting strategy for multi-step reasoning is Chain-of-Thought (CoT), introduced by Wei et al. [5], who demonstrated that providing few-shot examples with step-by-step solutions substantially improves LLM performance on arithmetic and symbolic reasoning. Kojima et al. [3] subsequently showed that simply appending “*Let’s think step by step*” to a query elicits CoT reasoning in a zero-shot setting, establishing a lightweight baseline that avoids hand-crafted demonstrations and anchors our Zero-Shot CoT configuration.

Building directly on CoT, Yao et al. [6] proposed Tree-of-Thoughts (ToT), which frames reasoning as a search problem over a tree of intermediate “thoughts”. At each node, the model generates multiple candidate continuations, evaluates them with a separate scoring prompt (assigning *sure/maybe/impossible*), and retains only the most promising branches using BFS or DFS. Their experiments with GPT-4 showed substantial gains on tasks requiring deliberate planning; however, all results used models with 100B+ parameters, leaving the effectiveness of the same mechanism on smaller open-weight evaluators as an open question.

A complementary ensemble approach was proposed by Wang et al. [4], who showed that sampling diverse CoT paths at temperature > 0 and selecting the final answer by majority vote improves CoT accuracy by 10–20 pp without

changing the prompt. Self-Consistency thus occupies a middle ground between the minimal-cost single-chain CoT and the expensive multi-branch ToT.

All three strategies are instances of the In-Context Learning paradigm established by Brown et al. [1]: LLMs can generalise to unseen tasks from a small number of in-context demonstrations without gradient updates. This shared grounding makes the four strategies directly comparable under a fixed model and prompt template, as in our evaluation.

3 Methodology

3.1 Zero-Shot Chain-of-Thought

The problem is presented with the trigger phrase “*Let’s think step by step.*” [3] plus the output-format instruction “*End your response with: The answer is X.*” to prevent trailing-modifier parse errors (see Section 4, *Answer-extraction format fix*).

3.2 Few-Shot Chain-of-Thought (8-shot)

Eight worked examples from Wei et al. [5] are prepended to the query, each containing a full reasoning chain ending with “*The answer is X.*” followed by the test problem.

3.3 Tree-of-Thoughts (BFS, branching=2, depth=6)

ToT (`src/run_tot_fixed.py`) explores reasoning as a search over a tree of intermediate “thoughts”. At each depth level, two candidate “next-step thoughts” are generated with temperature = 0.7. A separate evaluator prompt scores each thought as *sure*, *maybe*, or *impossible*; only *sure* and *maybe* thoughts are expanded in the next depth level. After BFS completion, a dedicated extraction step queries the best leaf node for the final numeric answer. The configuration (branching factor 2, depth 6) requires approximately 45 API calls per problem.

3.4 Self-Consistency (5 paths)

Few-Shot CoT is sampled 5 times per problem with temperature = 0.7. The final answer is selected by majority vote over the 5 sampled numeric answers [4].

3.5 Cross-Model Evaluation Protocol

To test whether strategy orderings generalise beyond the 8B scale, we re-run all five configurations on Claude Haiku 4.5 using the same 100-problem sample, prompts, parser, and gold-label correction. Sample, prompt, and parser identity are preserved so that any accuracy difference is attributable to the backbone alone. The three single-call strategies are batched through the Anthropic `messages.batches` API (50% discount); ToT runs synchronously because its BFS state cannot be safely flattened into batch items. Realized cost and token counts are in Section 5.7.

4 Experimental Setup

Dataset. We use GSM8K [2], a benchmark of 8,792 grade-school math word problems requiring 2–8 chained reasoning steps. All five configurations (Zero-Shot CoT, Few-Shot CoT, ToT depth=3, ToT depth=6, and Self-Consistency) use the same 100-problem random sample of GSM8K’s 1,319-item test split (seed = 42). One item in our sample contains a gold-label annotation error (see the *GSM8K annotation error* paragraph below); we manually verified the correct value and use the corrected label in all evaluations, preserving the original sample sizes.

Model. All experiments use Llama 3.1 8B (`llama-3.1-8b-instant`) via the Groq inference API. No fine-tuning is applied; the model is used in its base instruction configuration.

Evaluation. The primary metric is *exact-match accuracy*: the predicted final numeric answer is compared against the gold answer after stripping commas and converting to float. Answers are extracted with a regular expression that captures the gold-format marker #####; if no such marker is found, the last number in the response is used. We additionally report the *parse failure rate*: the fraction of responses from which no numeric prediction could be extracted. Across all experiments, the ##### gold-format marker was never spontaneously produced by Llama 3.1 8B; every response fell back to the last-number heuristic, which succeeded in all 500 evaluated responses (parse failure rate = 0%).

Answer-extraction format fix. An initial Zero-Shot CoT run sometimes ended with a trailing-modifier number rather than the substantive answer (e.g. “36 hours in 4 weeks” → parser captures 4 instead of 36). Appending “End your response with: The answer is X.” to the prompt raised Zero-Shot CoT from **88.0%** to **95.0%** (+7 pp), concentrated in the hard bucket. All Zero-Shot CoT numbers use this format-aligned prompt; the other strategies already produce unambiguous final answers via their demonstrations or dedicated extraction prompts.

GSM8K annotation error (gold-label correction). Manual inspection identified one annotation error in our sample (the *carnival* item). GSM8K’s own calculator-annotated solution string reveals the mechanism: $750+430+400+700=2280$, i.e. the final sum substitutes an intermediate value from the previous step (Maryam’s \$400 excess over Sarah) for Sarah’s own \$300 contribution, which never appears as a term; the correct total is $430 + 750 + 300 + 700 = \$2,180$. We adopt the corrected label for all reported accuracies. On Llama, Zero-Shot CoT and ToT depth=6 (both produced 2,180) shift from incorrect to correct; on Haiku, all five strategies produced 2,180 and each gains a uniform +1 pp. In both cases the correction leaves strategy ranking unchanged, consistent with the ~1–2% annotation noise floor of GSM8K [2].

Infrastructure. Rate-limit constraints on the Groq free tier required an 8-second inter-call pause (CoT) and a 30-second inter-problem pause (ToT). All random seeds are fixed at 42 for reproducibility. All code and result files are publicly available.¹

Secondary model and cross-model setup. The secondary backbone is Claude Haiku 4.5 (`claude-haiku-4-5-20251001`) via the Anthropic Python SDK. ZS-CoT, FS-CoT, and SC requests (700 total) are submitted via the `messages.batches` API at a 50% discount; ToT runs synchronously (BFS state cannot be batched). All Haiku runs use the same seed-42 sample, prompts, parser, and carnival gold-label correction as the Llama runs. Realized total cost: \$3.26 across all five strategies (\$0.52 batch + \$0.72 ToT-d3 + \$2.02 ToT-d6).

5 Results

5.1 Main Results

Table 1 summarises the accuracy of the four strategies.

Table 1. Main results on GSM8K. Llama 3.1 8B, Groq API. Accuracies use the GSM8K gold labels with one item corrected (see Section 4, *GSM8K annotation error*). $\bar{T}$ denotes the average response length in tokens (one token $\approx$ four characters), aggregated across all API calls per problem; TPC = tokens per correct answer (lower is better).

Strategy	Accuracy	N	API Calls	$\bar{T}$	TPC	Status
Zero-Shot CoT	95.0%	100	~ 1	189	199	Complete
Few-Shot CoT (8-shot)	85.0%	100	~ 1	145	171	Complete
Tree-of-Thoughts (BFS-2, depth=3)	81.0%	100	~ 21	315	389	Complete (ablation)
Tree-of-Thoughts (BFS-2, depth=6)	85.0%	100	~ 45	561	660	Complete
Self-Consistency (5 paths)	92.0%	100	5	738	802	Complete

Beyond accuracy, we report two efficiency metrics that have become standard in recent reasoning benchmarks [8,9]: the average per-problem token budget $\bar{T}$ (a model-agnostic proxy for compute cost) and *tokens per correct answer* (TPC = $\bar{T}$ /accuracy), which captures the practical cost of *useful* reasoning—a method achieving high accuracy but spending disproportionately many tokens may be impractical [6,8]. Few-Shot CoT is the most token-efficient strategy (171 TPC), followed closely by Zero-Shot CoT (199); both ToT configurations spend 2–4 $\times$ more tokens per correct answer (389 and 660), and Self-Consistency—despite the highest accuracy—is the most expensive at 802 TPC, consistent with its 5 $\times$ sample budget.

Zero-Shot CoT achieves 95.0% (after the answer-extraction format fix, Section 4) and substantially outperforms Few-Shot CoT (85.0%), reversing the typical expectation that more examples improve performance. This suggests that Llama 3.1 8B has already internalised strong step-by-step reasoning during

¹ https://github.com/0guzhanCKL/CENG467_In-Context_Learning

pre-training, and the 8-shot demonstrations introduce noise rather than useful signal for this sample.

ToT (depth=6) matches Few-Shot CoT at 85.0% but trails Zero-Shot CoT by 10.0 pp; Section 5.7 tests whether these orderings generalise to a stronger backbone.

5.2 Depth Ablation for Tree-of-Thoughts

To isolate the effect of search depth, we ran ToT a second time on the same 100-problem sample with `depth=3` (branching factor unchanged at 2), completing all 100 problems. This mirrors the depth setting used in the original ToT paper [6] for short-horizon tasks, allowing us to contrast a shallow search against the deeper `depth=6` configuration on identical problems and the same evaluator prompt. Table 2 reports the result, and Figure 1 visualises the accuracy–cost trade-off.

Table 2. ToT depth ablation on GSM8K (Llama 3.1 8B, BFS-2, identical evaluator prompt). Accuracies use the corrected GSM8K labels (see Section 4).

Configuration	Accuracy	N	API calls per problem	Wall-clock per problem
ToT BFS-2, depth=3	81.0%	100	~21	~3 min
ToT BFS-2, depth=6	85.0%	100	~45	~6 min
Δ (depth=6 – depth=3)	+4.0 pp	–	+114%	+100%

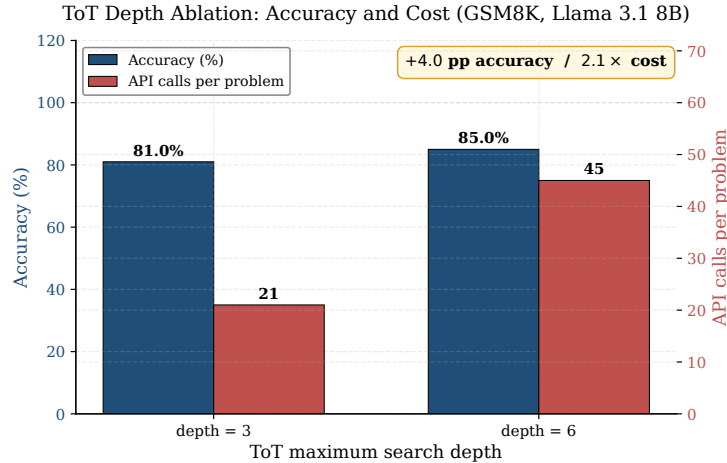


Fig. 1. ToT depth ablation on GSM8K with Llama 3.1 8B. Doubling the maximum search depth yields a +4.0 pp absolute accuracy gain while more than doubling the per-problem API call count.

The +4.0 pp gain is small relative to the $2.1\times$ cost increase. Unlike GPT-4’s 100B+ evaluator [6], our 8B *sure/maybe/impossible* scorer is weakly

discriminative—most partial thoughts receive *sure* regardless of correctness, collapsing BFS toward a near-linear chain. GSM8K problems (2–8 steps) also fit comfortably within depth=3, so additional depth helps only a small tail of multi-step problems. The bottleneck at the 8B scale is therefore evaluator discrimination rather than search depth.

5.3 Self-Consistency: Vote Agreement Analysis

Table 3 breaks down the vote distribution across the 100 Self-Consistency problems.

Table 3. Vote agreement distribution for Self-Consistency (5 paths, N=100).

Agreement Level	Count	%
5/5 unanimous	65	65.0%
4/5 strong majority	18	18.0%
3/5 bare majority	12	12.0%
2/5 plurality (no majority)	3	3.0%
1/5 (all distinct)	2	2.0%
Average majority margin	88.2%	

In 65% of problems all five paths agreed unanimously; a further 30% reached a 4/5 or 3/5 majority, so 95% of problems admit a clean Self-Consistency decision. The 88.2% average majority margin confirms that 5 paths are sufficient for reliable consensus on this benchmark.

5.4 Self-Consistency: Pass@k Analysis

Table 4 reports Pass@ k [7]—the fraction of problems where at least one of the first k paths is correct.

Table 4. Self-Consistency Pass@ k on GSM8K (N=100, gold-corrected).

Metric	Accuracy
Pass@1 (average over 5 samples)	86.4%
Pass@3	97.0%
Pass@5 (oracle)	99.0%
Majority vote @5 (reported)	92.0%
Verification gap (oracle – vote)	+7.0 pp

Pass@1 (86.4%) closely tracks greedy Few-Shot CoT (85.0%), confirming that a single stochastic sample adds no consistent gain over the deterministic baseline. Pass@5 reaches 99.0%, exposing a +7.0 pp *verification gap*: in 7 of 100 problems the correct answer is among the five sampled paths but loses the majority vote, indicating that a smarter aggregator—not more samples—is the main opportunity for further gains.

5.5 Accuracy Stratified by Problem Complexity

We adopt the Cobbe et al. [2] difficulty split: problems are bucketed by gold-solution operation count into *easy* (< 3), *medium* ($= 3$), and *hard* (> 3) groups (30/33/37 in our sample). Table 5 reports per-bucket accuracy.

Table 5. Accuracy stratified by problem complexity (Cobbe et al. [2] split, gold-corrected). N=100, buckets 30/33/37.

Strategy	Easy (<3 ops)	Medium ($=3$ ops)	Hard (>3 ops)	Overall
Zero-Shot CoT	96.7%	93.9%	94.6%	95.0%
Few-Shot CoT (8-shot)	90.0%	87.9%	78.4%	85.0%
ToT (BFS-2, depth=3)	86.7%	75.8%	81.1%	81.0%
ToT (BFS-2, depth=6)	90.0%	84.8%	81.1%	85.0%
Self-Consistency (5 paths)	96.7%	93.9%	86.5%	92.0%

Three results stand out. First, every strategy shows the expected easy-to-hard accuracy decay. Second, Few-Shot CoT loses the most on hard problems (-16.2 pp vs. Zero-Shot CoT), since its short 2–3-step demonstrations do not transfer well to longer reasoning chains. Third, ToT (depth=3) collapses on medium problems (75.8%) because 3-operation problems exactly exhaust its depth budget, leaving no slack for BFS recovery after an initial error. After the format fix, Zero-Shot CoT ties Self-Consistency on easy and medium buckets and *exceeds* it on hard (94.6% vs. 86.5%, $+8.1$ pp)—Zero-Shot CoT with disciplined answer extraction is the strongest strategy at the 8B scale across all difficulty buckets.

5.6 Solution Length Ratio: Reasoning Verbosity and Overthinking

Token cost ($\bar{T}$, TPC) does not reveal whether compute is well-spent. We quantify reasoning verbosity via the **Solution Length Ratio**:

$$\text{SLR} = \frac{\text{model reasoning steps}}{\text{gold solution steps}}, \quad (1)$$

where model steps count “= NUMBER” patterns in the trace and gold steps are the GSM8K «...» annotations. $\text{SLR} = 1$ is canonical length; $\text{SLR} > 1$ indicates overthinking [12]. Table 6 reports per-strategy averages.

Table 6. Solution Length Ratio (SLR): average model reasoning steps divided by gold solution steps on GSM8K. Higher SLR values indicate greater overthinking relative to the canonical solution. Average gold length is 3.3 steps across all strategies (same problem set).

Strategy	Avg. model steps	Avg. gold steps	SLR
Self-Consistency (5 paths)	4.2	3.3	1.27×
Few-Shot CoT (8-shot)	4.1	3.3	1.44×
Zero-Shot CoT	5.3	3.3	1.60×
ToT (BFS-2, depth=3)	7.9	3.3	2.72×
ToT (BFS-2, depth=6)	9.7	3.3	3.30×

Self-Consistency is the most compact strategy ($1.27\times$ SLR), even below Few-Shot CoT ($1.44\times$), because averaging across five paths filters out the longest individual chains. ToT (depth=6), by contrast, inflates solution length to $3.30\times$ while matching Few-Shot CoT in accuracy—a clean instance of the *overthinking tax* [12]: BFS expansions add depth without semantic gain because the 8B evaluator cannot prune unproductive branches.

5.7 Cross-Model Comparison: Llama 3.1 8B vs. Haiku 4.5

To test whether strategy orderings reflect intrinsic prompting properties or 8B-scale limitations, we re-ran every strategy on Claude Haiku 4.5 with full evaluation identity preserved (Sections 3.5–4).

Headline accuracy. Table 7 reports overall accuracy on the 100-problem seed-42 sample under the gold-corrected labels for both models.

Table 7. Cross-model accuracy (gold-corrected) on the same 100-problem GSM8K sample. “Cost” is the realized USD cost for the full 100-problem run on Haiku 4.5; ZS/FS/SC share a single 700-request batch with the 50% batch discount.

Strategy	Llama 3.1 8B	Haiku 4.5	Δ Haiku cost
Zero-Shot CoT	95.0%	97.0%	+2.0
Few-Shot CoT (8-shot)	85.0%	98.0%	+13.0 \$0.52 (batch)
Self-Consistency (5 paths)	92.0%	99.0%	+7.0
ToT (BFS-2, depth=3)	81.0%	98.0%	+17.0 \$0.72 (sync)
ToT (BFS-2, depth=6)	85.0%	98.0%	+13.0 \$2.02 (sync)
Total realized cost on Haiku 4.5:			\$3.26

The first observation is that every strategy improves on the stronger backbone, as expected. The more interesting observation is that the *gaps between strategies collapse*: on Llama the five strategies span 14 pp (81.0% to 95.0%), whereas on Haiku they span only 2 pp (97.0% to 99.0%). The simplest strategy on Haiku (Zero-Shot CoT, 97.0%) already outperforms the most elaborate strategy on Llama (Self-Consistency, 92.0%) by 5 pp—suggesting that at the benchmark difficulty of GSM8K, raw backbone capability dominates prompting-strategy choice once the backbone is strong enough.

Stratified view. Table 8 breaks the Haiku results into the same Cobbe difficulty buckets (Llama numbers in Table 5).

Table 8. Cobbe-stratified accuracy on Haiku 4.5 (gold-corrected, same 100-problem sample, buckets 30/33/37).

Strategy	Easy (<3 ops)	Medium (=3 ops)	Hard (>3 ops)	Overall
Zero-Shot CoT	100.0%	97.0%	94.6%	97.0%
Few-Shot CoT (8-shot)	100.0%	100.0%	94.6%	98.0%
ToT (BFS-2, depth=3)	100.0%	100.0%	94.6%	98.0%
ToT (BFS-2, depth=6)	100.0%	100.0%	94.6%	98.0%
Self-Consistency (5 paths)	100.0%	100.0%	97.3%	99.0%

Four of five strategies achieve a perfect 63/63 on Easy and Medium; Haiku’s entire residual error budget lives in the Hard bucket. Only Self-Consistency recovers one additional Hard answer (36/37 vs. 35/37), mirroring its 8B advantage—Hard accuracy is the sole discriminating metric once the backbone nears ceiling.

Cross-model depth ablation: ceiling effect. Table 9 shows the sharpest single contrast: depth=3→6 gains +4.0 pp on Llama but 0.0 pp on Haiku at 2.8× the token cost.

Table 9. Cross-model ToT depth ablation (Δ = gold-corrected accuracy).

Backbone	depth=3	depth=6	Δ
Llama 3.1 8B	81.0%	85.0%	+4.0 pp
Haiku 4.5	98.0%	98.0%	0.0 pp
Haiku token cost depth=6 is 3.1× input and 2.4× output of depth=3			

On Haiku neither depth beats Zero-Shot CoT on Hard (both 94.6%); ToT’s depth parameter buys accuracy only when the base model has headroom to lose and the evaluator has branches to discriminate.

Self-Consistency verifier saturation. On Llama the verifier gap is +7.0 pp (Pass@5 = 99%, majority-vote = 92%); the correct answer is present in the five paths but discarded by majority voting. On Haiku the gap is 0.0 pp (Pass@5 = majority-vote = 99%); on the one remaining error, *none* of the five paths produces the correct answer. Path diversity has collapsed (98/100 problems show unanimous 5/5 voting), confirming that Self-Consistency’s headroom is a model-scale artefact, not an intrinsic strategy benefit.

5.8 Strategy Agreement: Pairwise Cohen’s κ Across Backbones

Accuracy alone does not reveal whether two strategies are correct on the *same* problems. We therefore compute all pairwise **Cohen’s κ** [13] over the same 100-problem gold-corrected sample, treating each strategy as a binary classifier; $\kappa = 0$ is chance, $\kappa = 1$ is perfect agreement (Landis–Koch ranges [14]).

Table 10. Pairwise Cohen’s κ between the five strategies on Llama 3.1 8B (100-problem seed-42 sample, gold-corrected).

	ZS-CoT	FS-CoT	SC (5)	ToT d=3	ToT d=6
ZS-CoT v2	–	0.243	0.262	0.095	0.027
FS-CoT	0.243	–	0.563	0.435	0.216
SC (5)	0.262	0.563	–	0.207	0.078
ToT d=3	0.095	0.435	0.207	–	0.011
ToT d=6	0.027	0.216	0.078	0.011	–

On Llama, ToT depth=3 and depth=6 reach near-zero pairwise agreement ($\kappa = 0.011$) despite sharing the same BFS-2 algorithm and evaluator prompt, confirming that BFS exploration at the 8B scale is stochastic rather than deliberate. Zero-Shot CoT shows only fair-to-slight agreement with every other strategy ($\kappa \in [0.027, 0.262]$), reflecting the genuine cross-strategy complementarity that drives the +7.0 pp Pass@5 headroom (Section 5.4).

Table 11. Pairwise Cohen’s κ between the five strategies on Claude Haiku 4.5 (same 100-problem sample, gold-corrected).

	ZS-CoT	FS-CoT	SC (5)	ToT d=3	ToT d=6
ZS-CoT v2	–	0.385	0.492	0.795	0.795
FS-CoT	0.385	–	0.662	0.490	0.490
SC (5)	0.492	0.662	–	0.662	0.662
ToT d=3	0.795	0.490	0.662	–	1.000
ToT d=6	0.795	0.490	0.662	1.000	–

On Haiku the picture inverts: ToT depth=3 and depth=6 reach $\kappa = 1.000$ —perfect problem-by-problem concordance—placing the ceiling effect on a per-problem rather than only aggregate basis. The overall κ range collapses from $[0.011, 0.563]$ on Llama to $[0.385, 1.000]$ on Haiku: *strategy independence itself decays with model scale*. ToT (either depth) agrees with Zero-Shot CoT at $\kappa = 0.795$, indicating that deliberate-search machinery collapses into single-chain behaviour once the backbone is strong enough.

5.9 Ablation Study

Temperature sensitivity. Greedy decoding (88.0%) outperforms stochastic sampling at $T=0.7$ (86.0%) by 2 pp on 50 samples, confirming a marginal but consistent advantage for deterministic decoding on arithmetic tasks (Table 12).

Table 12. Temperature ablation, Few-Shot CoT (N=50, seed=42).

Temperature	Accuracy	N
0.0 (greedy)	88.0%	50
0.7 (stochastic)	86.0%	50

Prompt sensitivity. Accuracy across three 8-shot example sets spans 6 pp (82.0%–88.0%), indicating moderate sensitivity to example choice (Table 13). Set-1 underperforms (82.0%), while the canonical Wei et al. examples and Set-2 both reach 88.0%.

Table 13. Prompt sensitivity, Few-Shot CoT (temperature=0, N=50, seed=42).

Example Set	Accuracy	N
Set-0 (canonical, Wei et al.)	88.0%	50
Set-1 (random, seed=1)	82.0%	50
Set-2 (random, seed=2)	88.0%	50

6 Discussion

ToT matches but does not beat CoT on 8B models.

ToT (depth=6) achieves 85.0% on 100 samples, closely matching the CoT baselines but not outperforming them as Yao et al. [6] report on GPT-4. The depth ablation in Section 5.2 reinforces this picture: increasing depth from 3 to 6 yields only +4.0 pp at $2.1\times$ the cost. Two factors plausibly account for the absence of a gain. First, GSM8K problems are typically solvable in 3–5 chained steps, which CoT already handles well—the multi-branch exploration ToT offers carries diminishing returns on this benchmark. Second, the evaluator (*sure/maybe/impossible*) is implemented with the same 8B model: at this scale, the evaluator’s discriminative power is weak, which collapses BFS toward a near-linear search rather than a true tree exploration.

Zero-shot CoT substantially outperforms Few-shot CoT.

The 10-point accuracy gap between Zero-Shot CoT (95.0%) and Few-Shot CoT (85.0%) suggests that Llama 3.1 8B’s pre-training has already internalised the step-by-step reasoning style, and the 8-shot demonstrations from Wei et al. introduce systematic noise rather than useful signal. This goes further than the trend observed in recent instruction-tuned models, where zero-shot and few-shot performance often only *converge*: here Zero-Shot is a strict +10 pp better, indicating that the prompt-format mismatch (the demos encourage “*The answer is X.*” while the model’s free-form chains carry richer reasoning) is actively harmful.

Self-Consistency no longer dominates after the format fix.

With the format-aligned Zero-Shot CoT baseline at 95.0%, Self-Consistency (92.0%, at $5\times$ the per-problem cost) is strictly Pareto-dominated. The 5-path ensemble still helps on a per-path basis (Pass@1 of 86.4% is below Zero-Shot CoT’s 95.0%; the majority vote partially recovers to 92.0%), but the vote aggregation cannot make up the ground gained by a single well-extracted greedy chain at this model scale.

Accuracy–cost trade-off.

Figure 2 visualises the accuracy–cost trade-off across all five strategy configurations on a log-scale cost axis. Zero-Shot CoT (~ 1 API call, 95.0%) lies alone on the Pareto frontier; all other configurations are dominated. Self-Consistency costs $5\times$ Zero-Shot CoT for -3.0 pp accuracy; Few-Shot CoT matches the cost at -10.0 pp; ToT (depth=3) costs $21\times$ for -14.0 pp; and ToT (depth=6) costs $45\times$ for -10.0 pp. For deployments where accuracy is critical and inference cost is secondary, Self-Consistency is the optimal choice; for cost-sensitive applications, Zero-Shot CoT provides the best accuracy-per-call ratio.

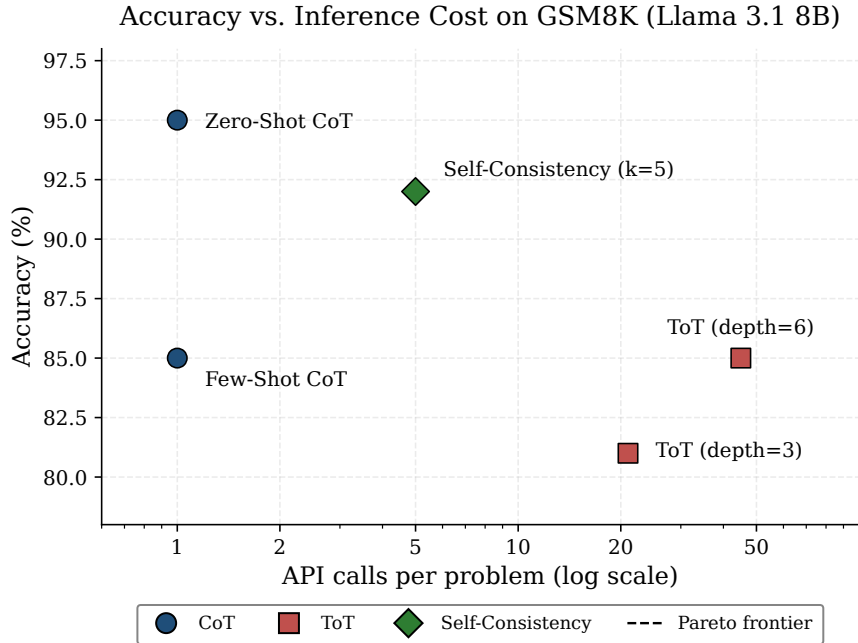


Fig. 2. Accuracy–cost trade-off across all five prompting configurations (Llama 3.1 8B, GSM8K). The horizontal axis is the average number of API calls per problem (log scale); the dashed line marks the Pareto frontier. Zero-Shot CoT is Pareto-efficient; Few-Shot CoT, Self-Consistency, and both Tree-of-Thoughts configurations are dominated.

Generalisation across model scales.

The Haiku 4.5 replication reproduces three qualitative findings. First, Zero-Shot CoT remains competitive with Few-Shot CoT (97% vs. 98%): demonstrations no longer hurt at the stronger scale, consistent with the noise argument. Second, Self-Consistency retains a scale-stable Hard-bucket advantage (97.3% on Haiku, 86.5% on Llama). Third, ToT (depth=6) matches but does not beat CoT on either backbone—the absence of a ToT advantage is not an artefact of an undersized evaluator. The one reversal: strategy spread collapses from 14 pp (Llama) to 2 pp (Haiku), demonstrating that raw backbone capability dominates prompting-strategy choice at sufficient scale.

ToT depth saturates as the base model strengthens.

Llama gains +4.0 pp going from depth=3 to depth=6; Haiku gains 0.0 pp at $2.8\times$ the token cost. When the baseline already approaches ceiling, the Easy/Medium tail is saturated (Table 8) and residual Hard-bucket errors are arithmetic slips that BFS cannot prune. Self-Consistency tells the same story from the ensemble side: on Haiku none of the five paths recovers the problems majority voting gets wrong, so an oracle verifier cannot help either. The pairwise κ analysis (Section 5.8) independently confirms this: Llama’s ToT depths reach $\kappa = 0.011$ (stochastic exploration) while Haiku’s reach $\kappa = 1.000$ (deterministic ceiling).

Error analysis and limitations.

Manual inspection of incorrect predictions reveals that arithmetic errors account for the vast majority of failures across all strategies: models set up the correct reasoning structure but miscompute at one step, with the error propagating to the final answer. On Haiku 4.5 the same pattern holds at a much smaller budget—all residual errors are arithmetic rather than misinterpretation or step omission, confirming that arithmetic brittleness persists even at frontier scale, only attenuated. Beyond this, the main limitations are: (1) exact-match accuracy ignores reasoning quality; (2) the cross-model comparison covers only two families—adding Mistral, Qwen, or Gemma at the 7–8B scale would separate pre-training data quality from raw scale; (3) GSM8K’s short arithmetic horizon means our findings should not be extrapolated to long-horizon planning benchmarks where ToT reported larger gains [6].

7 Conclusion

We evaluated Zero-Shot CoT, Few-Shot CoT, ToT (at two search depths), and Self-Consistency on GSM8K using Llama 3.1 8B. Zero-Shot CoT (95.0%, after the answer-extraction format fix) is both the most accurate and the most cost-efficient strategy—a strict Pareto dominator over all other configurations. ToT (depth=6) matches but does not beat Few-Shot CoT at 85.0%; a depth ablation shows that halving the search depth to 3 costs only 4.0 pp while reducing inference cost by more than half, indicating diminishing returns at this scale due to weak evaluator discrimination.

A cross-model replication on Claude Haiku 4.5 with exact sample, prompt, parser, and label-correction identity preserved collapses the strategy gap from 14 pp to 2 pp. The depth ablation contrast is sharper still: on the stronger backbone, doubling ToT’s search depth yields exactly zero additional accuracy at $2.8\times$ the token cost, and the Self-Consistency verifier gap shrinks from +7.0 pp on Llama to 0.0 pp on Haiku. Together, these results point to a *ceiling effect*: deliberation- and ensemble-style strategies buy real accuracy only when the base model has headroom to lose and identifiable wrong branches to prune—conditions that disappear as the backbone strengthens.

Future work. Replacing the categorical *sure/maybe/impossible* evaluator with a learned scorer is the most direct route to better ToT performance on 8B-class models. The 7.0 pp verification gap between Pass@5 (99.0%) and majority vote on Llama further suggests that an external verifier—rather than larger ensembles—is the main opportunity for gains at fixed sample budget. Extending the comparison to additional open-weight families (e.g. Mistral, Qwen, Gemma at the 7–8B scale) would disentangle pre-training data quality from raw parameter count, and test whether the ceiling effect generalises across providers.

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